Roll No. 124104069

NATIONAL INSTITUTE OF TECHNOLOGY, KURUKSHETRA
THEORY EXAMINATION
Question Paper

Month and Year of the Examination: December 2025
Programme: B.Tech. (Electrical Engineering)
Subject: Linear System Theory
Maximum Marks: 50

Date of Exam: 15/12/2025
Semester: Third (III)
Course No.: EEPC206
Time allowed: 3 Hours

Note: Attempt any **FIVE** questions in all. Each question is of 10 marks. If any data is missing or additional information is required, you may assume that same and write assumption in answer sheet.

Q.1. **(a)** Define a vector space and list its properties. **[3]**

(b) Consider the matrix $A = \begin{bmatrix} 0 & 1 \\ -3 & -4 \end{bmatrix}$. Linearly transform it into a diagonal matrix such that the diagonal matrix has same eigenvalues as that of this given matrix A. **[3]**

(c) Compute the column space and null space of the matrix $A = \begin{bmatrix} 1 & 0 & -3 & 2 \\ 0 & 1 & -5 & 4 \\ 3 & -2 & 1 & -2 \end{bmatrix}$ **[4]**

Q.2. **(a)** Prove that an orthogonal set of non-zero vectors is linearly independent. **[3]**

(b) Given a vector $v = [-10 \quad 4 \quad 3]^T$, compute its 1-norm, 2-norm, and infinite-norm. **[3]**

(c) Prove that a $n \times n$ matrix A is diagonalizable if and only if A has n linearly independent eigenvectors. **[4]**

OR

(c) Find the 1-norm, 2-norm, and infinite-norm of the matrix $A = \begin{bmatrix} 0 & 1 \\ -9 & -20 \end{bmatrix}$. Also compute its Frobenius norm. **[4]**

Q.3. **(a)** Explain the Gram-Schmidt orthogonalization process, and compute the orthogonal basis for the following set of vectors $\begin{bmatrix} 0 \\ 2 \\ 4 \end{bmatrix}, \begin{bmatrix} 5 \\ -7 \\ 6 \end{bmatrix}$ **[4]**

(b) Make a change of variable $x = Py$, where $x = \begin{bmatrix} x_1 \\ x_2 \end{bmatrix}$ and $y = \begin{bmatrix} y_1 \\ y_2 \end{bmatrix}$ that transforms the quadratic form $3x_1^2 - 2x_2^2 + 4x_1x_2$ into a quadratic form with no cross product term. **[3]**

(c) State and explain the singular value decomposition and its applications. What are singular values, left and right eigenvectors of a matrix. **[3]**

Q.4. **(a)** State and explain Cayley Hamilton theorem, and apply it to obtain state transition matrix $\phi(t)$ for a LTI system $\dot{x} = Ax + Bu$. **[4]**

(b) Obtain the state variable model of the network shown in Fig. 1. Consider x_1 and x_2 as shown in diagram as state variables, u and y as shown as the system input and output variables. x_3

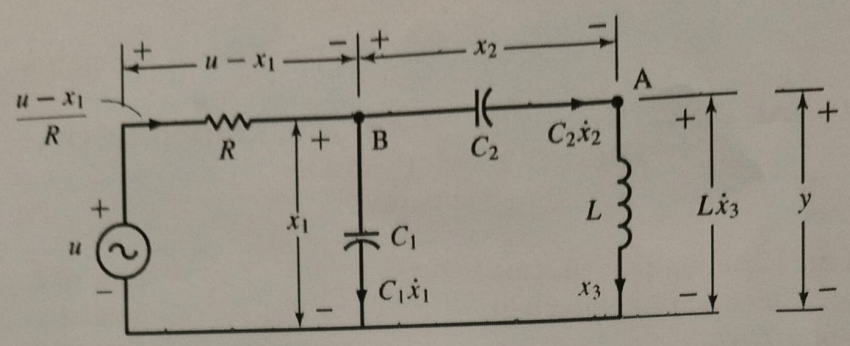
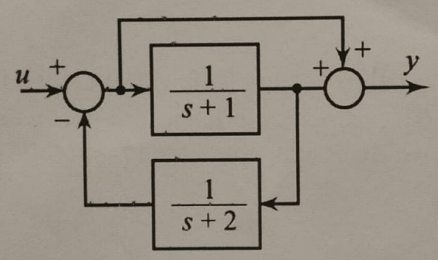


Figure 1

Q.5. **(a)** Discuss the need for linearizing engineering systems. Explain the linearization of nonlinear system using first-order Taylor Series. **[3]**

(b) Prove that a LTI autonomous system $\dot{\boldsymbol{x}} = \boldsymbol{Ax}$ is stable in the sense of Lyapunov if there exists a positive definite matrix $\boldsymbol{P}$ such that $\boldsymbol{A}^T\boldsymbol{P} + \boldsymbol{PA} = -\boldsymbol{Q}$, where $\boldsymbol{Q}$ may be positive definite or positive semidefinite matrix. **[3]**

(b) Convert the block diagram given below to state diagram by converting each block into equivalent controllable canonical form and then find the overall state space representation. Show that the system is stable in sense of Lyapunov. **[4]**



Q.6. **(a)** Given a state space representation $\dot{\boldsymbol{x}} = \boldsymbol{Ax} + \boldsymbol{Bu}$ $\boldsymbol{y} = \boldsymbol{Cx} + \boldsymbol{Du}$. Derive expression for its equivalent transfer function. Further, show that all state space representations of a given system refer to same transfer function. **[5]**

(b) A regulator system has the plant **[5]**

$$\dot{x} = \begin{bmatrix} 0 & 1 \\ 20 & 0 \end{bmatrix} x + \begin{bmatrix} 0 \\ 1 \end{bmatrix} u \qquad y = [1 \quad 0]x$$

Design a control law $\boldsymbol{u} = -\boldsymbol{Kx}$, so that the closed-loop system has eigenvalues at $-1 \pm j2$

Q.7. **(a)** Define controllability and observability and how will you ascertain them for a LTI system. **[4]**

(b) Consider the system **[6]**

$$\dot{x} = \begin{bmatrix} -2 & -1 \\ 1 & 0 \end{bmatrix} x + \begin{bmatrix} 0 \\ 1 \end{bmatrix} u \qquad y = [0 \quad 1]x$$

Design a full order state estimator, so that observer poles located at $-10, -20$. Explain why we place eigenvalues of an estimator far away from imaginary axis.

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